



Lecture 11 — Topper Notes

Complete Structured Notes | All 28 Questions & Concepts Covered

Biology · Sports · Geography · Environment · Economy · Polity · History · Art & Culture



① Biology — Carbohydrates & Sugars

Q1. What are oligosaccharides?

- ☐ A Carbohydrates that yield one monosaccharide unit on hydrolysis
- ☐ B Carbohydrates that yield more than 10 monosaccharide units on hydrolysis
- ☒ C Carbohydrates that yield 2 to 10 monosaccharide units on hydrolysis ✓
- ☐ D Carbohydrates that do not yield any monosaccharide units on hydrolysis

✓ Answer: C — Oligosaccharides yield 2-10 monosaccharide units on hydrolysis

KEY FACTS

- 🔑 **Monosaccharide** = yields 1 unit on hydrolysis
- 🔑 **Oligosaccharide** = yields 2-10 units on hydrolysis
- 🔑 **Polysaccharide** = yields more than 10 units on hydrolysis

→ Common Sugars — Source-wise

Sugar	Found In	Note
Maltose	Cereals & grains	Disaccharide of glucose + glucose
Lactose	Milk	"Milk sugar" — many people are lactose intolerant
Fructose	Fruits	"Fruit sugar" — sweetest natural sugar

QUICK FACT

Protein present in our **hair and nails** = **Keratin** (a fibrous structural protein)



② Sports — Hockey & India's Olympic Legacy 🏑

Q2. Which statement about the goal post in hockey is correct?

☒ A Height = 2.14 m, Width = 3.66 m ✓

☐ B Height = 3.66 m, Width = 2.14 m

☐ C Height = 2.44 m, Width = 3.66 m

☐ D Height = 2.14 m, Width = 3.05 m

✓ Answer: A — Hockey goal post: Height 2.14 m, Width 3.66 m | 11 players per side on field

INDIA — HOCKEY OLYMPIC LEGACY

6 consecutive Gold medals: 1928 → 1956

Gold again in 1964

Last Gold won in 1980 (Moscow)

Recent 2 Olympics (Tokyo 2020 & Paris 2024) → **Bronze, back-to-back**



③ Environment — Kuno National Park & Project Tiger 🐅

Q3. Kuno National Park is primarily famous for which programme?

☐ A Project Elephant

☐ B Rhino Conservation Programme

☒ C Cheetah Reintroduction Programme ✓

☐ D Vulture Conservation Programme

✓ Answer: C — Kuno NP is famous for the Cheetah Reintroduction Programme (African cheetahs brought from South Africa & Botswana)

KEY FACTS

- 🔑 Kuno National Park is located in **Sheopur district, Madhya Pradesh** (Gwalior–Shivpuri region), on the banks of the **Kuno River**
- 🔑 Cheetah's IUCN status = **Vulnerable**
- 🔑 Cheetah was declared **extirpated (locally extinct) from India in 1952**
- 🔑 **Asiatic cheetah** ≠ **African cheetah** — different species; the African cheetah is faster but has a smaller mouth
- 🔑 India reintroduced **African cheetahs from South Africa & Botswana** into Kuno NP under the **Cheetah Reintroduction Programme**

→ Project Tiger — Quick Facts

PROJECT TIGER

Launched in **1973**
First Tiger Reserve = **Jim Corbett National Park**
Jim Corbett NP = India's **first National Park**, established **1936**

MP NATIONAL PARKS — REVISION

Kanha National Park → Madhya Pradesh
Dudhwa National Park → Uttar Pradesh (Lakhimpur Kheri)
Kuno National Park → Madhya Pradesh



④ Environment — Madhav National Park (Current Affairs) 🐅

CURRENT AFFAIRS

- 🔑 **Madhav National Park, Shivpuri (MP)** was declared India's **58th Tiger Reserve** — the **9th in Madhya Pradesh**
- 🔑 This is why it was recently in the news
- 🔑 MP continues to be India's **"Tiger State"** with the most tiger reserves of any state

MEMORY TRICK

Mnemonic: every time a new "X National Park becomes a Tiger Reserve" current-affairs question shows up — first check (1) which number reserve it is nationally, and (2) which number it is within its own state.



⑤ Geography — Peninsular Plateau vs Himalayan Region

Q5. How is the Peninsular Plateau of India different from the Himalayan region in terms of geological stability?

- ☒ A Peninsular Plateau is ancient & stable; Himalayas are young & still changing ✓
- ☐ B Peninsular Plateau is young & unstable; Himalayas are ancient & stable
- ☐ C Both are equally old and stable
- ☐ D Both are equally young and unstable

✓ Answer: A — Peninsular Plateau = old, stable shield (low earthquake risk); Himalayas = young fold mountains, still rising (active convergent boundary → high earthquake & landslide risk)

→ Geomorphic Processes — Endogenic vs Exogenic

Force	Meaning	Includes	Collective Term
Endogenic	Internal forces (plate tectonics)	Orogenic + Epeirogenic + Volcanism + Earthquakes	Diastrophism
Exogenic	External forces, energy from the Sun	Weathering, Erosion, Mass Movement	Denudation

OROGENIC vs EPEIROGENIC

Orogenic ("oro" = mountain) → mountain-building processes

Epeirogenic → slow upliftment/subsidence of large parts of Earth's crust

EXOGENIC — DRIVING FORCES

Weathering → molecular stresses

Erosion → kinetic energy

Mass movement → gravity

⚠ Weathering is **NOT** a prerequisite for mass movement

→ Plate Tectonics — Who Said What?

- ☐ 1912 **Alfred Wegener** — Continental Drift Theory (first to propose that continents were once joined)
- ☐ 1930s **Arthur Holmes** — Convection Current Theory (explains why plates move)
- ☐ 1960s **Harry Hess** — Sea Floor Spreading Theory
- ☐ 1967–68 **McKenzie, Parker & Morgan** — formalised the modern Plate Tectonic Theory

⚠ EXAM TRAP!

Don't confuse: **Wegener** = continents drift (1912) ≠ **Holmes** = explains why, via convection currents (1930) ≠ **McKenzie/Parker/Morgan** = formal Plate Tectonics Theory (1967–68)

KEY FACTS

- 3 types of plate boundary movement:
- Convergent** = Destructive
- Divergent** = Constructive
- Transform** = Conservative (neither constructive nor destructive)
- Indian Plate & Eurasian Plate boundary = **Convergent** & still active → Himalayas still rising



⑥ Geography — Trans-Himalayas & Himalayan Divisions

→ Trans-Himalayas — 3 Ranges

KEY FACTS

- Karakoram, Ladakh, Zaskar** ranges form the Trans-Himalayas
- Karakoram has glaciers: **Siachen, Baltoro, Biafo, Hispar**
- K2 (Godwin Austen)** — height **8,611 m** — 2nd highest peak in the world & India's highest peak, but located in **PoK**

→ Himalaya — Three Parallel Ranges

Range	Also Called	Avg. Height	Key Features
Great Himalaya	Himadri / Inner Himalaya	6,000 m	Mt. Everest (8,848 m), Kanchenjunga (8,598 m, Sikkim), Kashmir Valley
Lesser/Middle Himalaya	Himachal / Middle Himalaya	4,000 m	Pir Panjal (Kashmir), Dhauladhar (HP), Nag Tibba (Uttarakhand), Mahabharat Range (Nepal)
Shivalik	Outer Himalaya	~900–1,100 m	Disappears in the North-East, replaced by Duars

DUNS & KAREWAS

Duns = longitudinal valleys between Lesser Himalaya & Shivalik

E.g. **Dehradun** (largest), Patlidun, Kotlidun

Karewas = lacustrine/glacial deposits in Kashmir Valley → ideal for **saffron** cultivation

HIGH PEAKS

Mt. Everest — 8,848 m (Great Himalaya)

Kanchenjunga — 8,598 m (Sikkim) — 3rd highest in world

K2/Godwin Austen — 8,611 m (PoK)



⑦ Economy — Global Hunger Index (GHI) 2025

CURRENT AFFAIRS

🔑 India ranks **102nd out of 123 countries** in GHI 2025, with a score of **25.8** — "Serious" hunger category

🔑 Published jointly by **Concern Worldwide** (Ireland) & **Welthungerhilfe** (Germany)

🔑 4 indicators: **Undernourishment, Child Stunting, Child Wasting, Child Mortality**

🔑 KEY FACTS

🔑 **Stunting** = low height-for-age (chronic malnutrition)

🔑 **Wasting** = low weight-for-height (acute malnutrition)

🔑 **Child mortality** = death rate of children under 5

Note: India's 2024 GHI rank was 105th/127 with score 27.3 — improved to 102nd/123 (25.8) in 2025.



⑧ Economy — HDI vs MPI vs National MPI

Index	Released By	Dimensions	Indicators
HDI (Human Development Index)	UNDP	Long & healthy life, Knowledge, Decent standard of living	Life expectancy; Expected + Mean years of schooling; GNI per capita

Global MPI (Multidimensional Poverty Index)	UNDP + OPHI	Health, Education, Standard of living	10 indicators
National MPI (India)	NITI Aayog	Same 3 dimensions	12 indicators (adds bank account & maternal/prenatal care)

🧠 MEMORY TRICK

Remember Global MPI's 3 dimensions with **"SHE"** → Standard of living, Health, Education. You don't need to memorise the dimension index formula — just the dimensions & indicators.



⑨ Economy — Types of Unemployment

Q? Unemployment that occurs due to regular seasonal changes in a year is called?

- ☐ A Frictional Unemployment
- ☐ B Structural Unemployment
- ☒ C Seasonal Unemployment ✓
- ☐ D Cyclical Unemployment

✓ Answer: C — Seasonal unemployment is mainly seen in India's agriculture sector (idle between sowing & harvesting seasons)

Type	Cause	Example/Note
Frictional	When someone switches from one job to another	Also called search/voluntary unemployment
Structural	Skills gap — worker's skill ≠ market demand	—
Cyclical	Recession-related	Jobs lost in downturn, regained as economy recovers
Seasonal	Regular seasonal changes in a year	India's agriculture — idle outside sowing/harvest season
Disguised	Hidden — workers present but productivity = zero	Common in Indian agriculture (too many people on small land)
Educated	Has a degree but no job	—

⑩ Economy — Money Supply: M0 to M4 💰

Q10. Which of the following is known as Narrow Money?

☒ A M1 ✓

☐ B M2

☐ C M3

☐ D M4

✓ Answer: A — M1 = Narrow Money | M3 = Broad Money / Monetary Aggregate

Aggregate	Composition	Note
M0	Currency in circulation + Deposits with RBI (bank + other deposits)	Called High-Powered Money / Base Money / Reserve Money
M1	Currency with public + Demand deposits with banks + Other deposits with RBI	Narrow Money
M2	M1 + Post office savings deposits	—
M3	M1 + Time/Term Deposits with banks	Broad Money / Monetary Aggregate
M4	M3 + Total Post Office deposits	—

🧠 MEMORY TRICK

Bank deposits are **liabilities** for banks — split into **Demand** liabilities (savings, current a/c) and **Term/Time** liabilities (FDs, RDs). M1 takes demand deposits; M3 adds time deposits.

⑪ Schemes — INSPIRE Programme (DST) 🔬

Q11. Under the Scholarship for Higher Education (SHE) component of the INSPIRE programme, what is the annual scholarship amount provided to selected students?

- ☐ A ₹10,000 per year
- ☐ B ₹50,000 per year
- ☒ C ₹80,000 per year ✓
- ☐ D ₹1,00,000 per year

✓ Answer: C — ₹80,000 per year for up to 5 years + mentorship support

KEY FACTS

- 🔑 **INSPIRE** = Innovation in Science Pursuit for Inspired Research — initiative of the Department of Science & Technology (DST)
- 🔑 **INSPIRE Award – MANAK**: for students of **Class 6–10** (age 10–15) → grant of ~₹10,000 for an innovative project for district/state exhibitions
- 🔑 **SHE (Scholarship for Higher Education)**: for UG/PG students (age 17–22) → ₹80,000/year for up to 5 years + mentorship
- 🔑 **AORC (Assured Opportunity for Research Career)**: for PhD/post-doctoral researchers (age 22–32) → assured research opportunity + financial assistance



12 Polity & History — Land Reforms Post-1947 🌾

Q12. What was a major goal of the land reforms policy introduced after 1947?

- ☐ A Selling land to foreign companies
- ☒ B Abolishing the Zamindari system ✓
- ☐ C Converting all farmland into industrial land
- ☐ D Privatising irrigation systems

✓ Answer: B — Abolition of the Zamindari system was a core goal of post-Independence land reform

→ Colonial Land Revenue Systems — Quick Revision

System	Introduced By	Year	Region
Permanent (Zamindari) Settlement	Lord Cornwallis	1793	Bengal, Bihar, Odisha

Ryotwari System	Thomas Munro	1820	Madras (based on Ricardian Theory of Rent)
Mahalwari System	Holt Mackenzie & William Bentinck	—	Punjab & North-West Frontier region

KEY FACTS

- Core policy: **"Land to the Tiller"** — actual cultivator gets ownership right
- Land Ceiling Act** capped the maximum land a person/family could hold; surplus land redistributed
- Big landowners challenged this citing **Right to Property** as a Fundamental Right
- 44th Amendment (1978)** removed Right to Property from Fundamental Rights, made it a legal right under Article 300A, Part XII
- 1st Amendment** added the **9th Schedule** + Articles **31A & 31B** — laws in the 9th Schedule were shielded from judicial review

EXAM TRAP!

I.R. Coelho case (2007): Supreme Court ruled that laws added to the 9th Schedule after the **Kesavananda Bharati case (24 April 1973)** — which laid down the **Basic Structure Doctrine** — CAN be reviewed by courts if they violate the basic structure.



⑬ Science & Geography — BARC & Nuclear Power Plants

Q13. **BARC (Bhabha Atomic Research Centre)** is located in which city?

- ☒ A **Trombay, Mumbai** ✓
- ☐ B ~~Kalpakkam, Tamil Nadu~~
- ☐ C ~~Vadodara, Gujarat~~
- ☐ D ~~Indore, Madhya Pradesh~~

✓ Answer: A — BARC is at Trombay, Mumbai

Nuclear Power Plant	State	Note
Tarapur	Maharashtra	India's first NPP — 1969
Kalpakkam, Kudankulam	Tamil Nadu	—

Narora	Uttar Pradesh	—
Kakrapar	Gujarat	—
Kaiga	Karnataka	—
Rawatbhata	Rajasthan	—



14 Current Affairs — Tangail Sarees GI Tag

Q14. Tangail Sarees, which received a GI tag in 2024, are a traditional handloom product of which state?

- ☐ A Odisha
- ☒ B West Bengal ✓
- ☐ C Assam
- ☐ D Bihar

✓ Answer: B — "Tangail Saree of Bengal" got the GI tag in January 2024, woven mainly in Nadia & Purba Bardhaman districts

BENGALI NEW YEAR

The first day of the Bengali month **Boishakh** is celebrated as the New Year — **Poila Boishakh / Pohela Boishakh**



15 International Relations — Quad vs NATO

Q15. What is unique about the institutional status of the Quad compared to NATO?

- ☐ A Quad has a permanent headquarters like NATO
- ☐ B Quad is a treaty-bound military alliance
- ☒ C Quad is informal diplomacy, not a formal treaty-based alliance ✓
- ☐ D Quad has a collective defence clause like NATO's Article 5

✓ Answer: C — Unlike NATO, the Quad is informal strategic diplomacy with no binding treaty and no permanent headquarters

Feature	NATO	Quad
Full Form	North Atlantic Treaty Organization	Quadrilateral Security Dialogue
Members	32 countries (N. America + Europe)	India, USA, Japan, Australia
Nature	Formal treaty-based military alliance	Informal strategic grouping
Key clause	Article 5 — attack on one = attack on all	No such binding clause
HQ	Brussels, Belgium	No permanent HQ
Joint Exercise	—	Malabar Exercise

RUSSIA—UKRAINE CONTEXT

Ukraine sought a **NATO membership guarantee** (though never formally joined) — Russia saw this as a security threat (NATO expanding to its doorstep), a key trigger behind the ongoing conflict.



⑩ Polity — Articles 73–85 (Union Executive) ⚖️

Q16. Which article deals with the extent of executive power of the Union, defining the scope and limitations of the Union government's authority in administration?

☒ A Article 73 ✓

☐ B Article 74

☐ C Article 75

☐ D Article 77

✓ Answer: A — Article 73 = extent of executive power of the Union

Article	Subject
73	Extent of executive power of the Union
74	Council of Ministers headed by the PM, to aid & advise the President
75	PM & other Ministers — appointment, tenure, oath, salary (6 clauses)
76	Attorney General of India
77	Conduct of business of Govt of India — in the name of the President
78	Duties of the Prime Minister

79	Constitution of Parliament
80	Composition of the Rajya Sabha
81	Composition of the Lok Sabha
82	Delimitation after each census
83	Duration of the Houses of Parliament
84	Qualifications for membership of Parliament
85	Sessions, prorogation & dissolution of Parliament

🧠 MEMORY TRICK

Union Article + 89 = Corresponding State Article (e.g., 73 → 162, 74 → 163, 75 → 164, 77 → 166). Handy shortcut when you forget the State-list equivalent!

→ Article 75 — Six Clauses

🔑 KEY FACTS

- 🔑 (1) President appoints PM; other Ministers appointed on PM's advice
- 🔑 Ministers hold office **during the pleasure of the President** (individual responsibility to President; collective responsibility to Lok Sabha)
- 🔑 (4) Oath administered by the President (or a person he appoints, e.g. the **Pro-tem Speaker**)
- 🔑 (5) A Minister who is not a member of either House for **6 consecutive months** loses ministership
- 🔑 (6) Salaries & allowances of Ministers are determined by **Parliament**

PRO-TEM SPEAKER

A temporary Speaker, appointed by the **President** (usually the senior-most member of the new Lok Sabha), whose job is to administer the oath to newly elected MPs — until a permanent Speaker is elected.

ARTICLE 74 — BINDING ADVICE

42nd Amendment — made President's reliance on Council of Ministers' advice

binding

44th Amendment — President may return advice **once** for reconsideration; reconsidered advice is binding

PRESIDENT vs PM

President = **de jure / nominal / titular** head

PM = **de facto** head, real executive power



⑰ Modern History — Government of India Acts

Q17. Which of the following acts proposed the creation of Public Service Commissions for both the Federation and individual provinces?

- ☐ A ~~Government of India Act, 1919~~
- ☒ B **Government of India Act, 1935** ✓
- ☐ C ~~Indian Councils Act, 1909~~
- ☐ D ~~Indian Independence Act, 1947~~

✓ Answer: B — The 1935 Act called for a Federal Public Service Commission

Act	Also Known As	Key Provisions
Gol Act 1858	Queen's Proclamation	Ended Company rule; created Secretary of State for India, assisted by a 15-member Council of India
Indian Councils Act 1909	Morley-Minto Reforms	Supplementary questions allowed on budget; Satyendra Prasad Sinha — first Indian on Viceroy's Executive Council; Council strength 16 → 60; separate electorates for Muslims introduced (Minto = "Father of Communalism")
Gol Act 1919	Montagu-Chelmsford / Montford Reforms	Introduced Dyarchy in provinces (Reserved vs Transferred subjects); Bicameral legislature at Centre; extended separate electorates to Sikhs, Anglo-Indians & Christians
Gol Act 1935	—	Abolished provincial dyarchy → Provincial Autonomy ; introduced dyarchy at the Centre; Bicameral legislature in 6 of 11 provinces; proposed All India Federation (never formed) & Federal Public Service Commission
Indian Independence Act 1947	Based on Mountbatten Plan	Partitioned India into two dominions — India & Pakistan

1892 vs 1909

Power of **discussion** on the budget → Indian Councils Act **1892**

Power to ask **supplementary questions** → 1909 Act



18 Geography — Tropical Thorn Forest & Vegetation Types

Q18. Which of the following trees is primarily found in the tropical thorn forest?

- ☐ A ~~Banyan and Peepal~~
- ☒ B **Khejri and Ber** ✓
- ☐ C ~~Mahua and Amla~~
- ☐ D ~~Pine and Juniper~~

✓ Answer: B — Khejri & Ber are typical thorn-forest (xerophytic) trees

Forest Type	Rainfall	Trees
Coniferous (Himalayan)	—	Pine, Cedar, Deodar (cone-shaped, gymnosperms)
Moist Deciduous	150–200 cm	Mahua, Amla
Dry Deciduous	70–150 cm	Palash, Amaltas, Tendu (known for "Parkland landscapes")
Tropical Thorn Forest	< 70 cm (some areas < 50 cm)	Khejri, Ber

KEY FACTS

🔑 Tropical Deciduous Forest = "**Monsoon Forest**" — largest forest cover in India; split into Moist & Dry deciduous

🔑 Thorn forest trees = **xerophytic** (adapted to dry conditions); leaves modified into thorns

PLANT ADAPTATION TYPES

Xerophytes → adapted to dry conditions

Heliophytes → sun-loving plants

Halophytes → salt-loving plants (e.g., mangroves, Sundarbans)

Neophytes → non-native introduced species (e.g., tobacco, potato — brought by the Portuguese)

GYMNOSPERMS vs ANGIOSPERMS

Gymnosperms (conifers) = naked seeds, non-flowering

Angiosperms = flowering plants



19 Static GK — City Nicknames of India

Q19. Which city is known as the "Brass City of India" for its brass handicrafts industry?

- ☐ A Jaipur
- ☒ B Moradabad ✓
- ☐ C Jodhpur
- ☐ D Aligarh

✓ Answer: B — Moradabad = Brass City of India

City	Nickname
Jaipur	Pink City
Udaipur	City of Lakes
Moradabad	Brass City of India
Aligarh	City of Locks
Mirzapur	Famous for Carpets
Ahmedabad	Manchester of India (earlier this title belonged to Kanpur)
Jamshedpur	Steel City of India



20 Economy — SEBI & SIDBI

KEY FACTS

- 🔑 **SEBI** = Securities and Exchange Board of India — established **1988**; received **statutory recognition in 1992**
- 🔑 Regulates securities — mutual funds, stock markets, etc.
- 🔑 Current Chairman (as of this lecture): **Tuhin Kanta Pandey** (11th SEBI Chairman, since March 2025)

KEY FACTS

- 🔑 **SIDBI** = Small Industries Development Bank of India — established **1990**
- 🔑 Headquarters: **Lucknow**



21 History/GK — Mehrangarh Fort, Jodhpur 🏰

Q21. Which famous fort is located in Jodhpur and was built by Rao Jodha?

- ☐ A Amer Fort
- ☒ B Mehrangarh Fort ✓
- ☐ C Chittorgarh Fort
- ☐ D Kumbhalgarh Fort

✓ Answer: B — Mehrangarh Fort, Jodhpur — built by Rao Jodha, a Rathore ruler



22 Polity — Objectives Resolution & Constituent Assembly 🇮🇳

Q22. The Objectives Resolution was moved in the Constituent Assembly by:

- ☒ A Jawaharlal Nehru ✓
- ☐ B Sardar Vallabhbhai Patel
- ☐ C B.R. Ambedkar
- ☐ D Rajendra Prasad

✓ Answer: A — Nehru moved the Objectives Resolution on 13 December 1946; adopted 22 January 1947. It forms the basis of the Preamble (not the Preamble itself)

→ Constituent Assembly — Formation

- ☐ **1946** Formed per the **Cabinet Mission Plan** (members: Pethick-Lawrence, Stafford Cripps, A.V. Alexander; Lawrence = Chairman)
- ☐ — Proposed 389 members → 296 from British India (292 from 11 Governor's Provinces + 4 from Chief Commissioner's Provinces) + 93 nominated from Princely States. Partially elected (indirectly), partially nominated
- ☐ **9 Dec 1946** First meeting — 211 members attended (Muslim League boycotted); after Partition, 299 seats remained
- ☐ — **Sachchidananda Sinha** = temporary Chairman → **Rajendra Prasad** = permanent President; H.C. Mukherjee & V.T. Krishnamachari = Vice-Presidents; **B.N. Rau** = Constitutional Advisor

- **13 Dec 1946** Nehru moves the **Objectives Resolution**
- **22 Jan 1947** Objectives Resolution **adopted**
- **May 1949** Commonwealth membership ratified
- **22 Jul 1947** National Flag adopted
- **24 Jan 1950** National Anthem & National Song adopted; Rajendra Prasad elected first President of India

CONSTITUENT ASSEMBLY AS LAW-MAKER

The Constituent Assembly also functioned as a **Provisional Parliament**, presided over by **G.V. Mavalankar**

→ Preamble – Key Points

KEY FACTS

- 🔑 Source of power: **"We, the People of India"**
- 🔑 Nature: Sovereign, Socialist, Secular, Democratic, Republic
- 🔑 Objectives: **Justice** (social, economic, political — 3 types), **Liberty** (thought, expression, belief, faith, worship — 5 types), **Equality** (status & opportunity — 2 types), **Fraternity** (1 type)
- 🔑 **Liberty, Equality, Fraternity** → borrowed from the **French Revolution**
- 🔑 **Justice** (social/economic/political) → inspired by the **Russian Revolution**
- 🔑 Date of adoption: **26 Nov 1949** (Constitution Day) | Came into force: **26 Jan 1950** (Republic Day)
- 🔑 Provisions on citizenship, elections & the provisional parliament came into effect from **26 Nov 1949** itself

→ Key Committees of the Constituent Assembly

Committee	Head
Union Powers Committee	Jawaharlal Nehru
Advisory Committee on Fundamental Rights, Minorities & Tribal Areas	Sardar Vallabhbhai Patel
Drafting Committee (7 members, formed 29 Aug 1947)	B.R. Ambedkar — "Father of the Constitution"
Steering / Rules of Procedure Committee	Rajendra Prasad

Any committee whose name has **"Union"** in it → headed by **Nehru**. Sardar Patel = "Iron Man / Bismarck of India."



23 Polity — Writs (Articles 32 & 226)

Q23. Which writ examines the legality of a person holding a public office?

- ☐ A Habeas Corpus
- ☐ B Mandamus
- ☒ C Quo Warranto ✓
- ☐ D Prohibition

✓ Answer: C — Quo Warranto = "By what authority" — issued when a person holds a public office they are not legally entitled to

Writ	Meaning	When Issued
Habeas Corpus	"To have the body of"	Person unlawfully detained
Mandamus	"We command"	Public official fails to perform a mandatory (not discretionary) duty; not against President/Governor
Certiorari	"To be certified"	Higher court transfers/quashes a case where a lower court exceeded its jurisdiction
Prohibition	"To forbid"	Higher court stops a lower court from exceeding its jurisdiction (preventive)
Quo Warranto	"By what authority"	Person holding a public office without legal entitlement

- Writs are a concept borrowed from the **UK/British Constitution** (along with martial law, office of CAG, bicameral legislature, parliamentary form of govt.)
- Supreme Court** issues writs under **Article 32** — itself a Fundamental Right, so the SC cannot refuse to issue one
- High Court** issues writs under **Article 226** — not a Fundamental Right, but wider in scope since it covers both Fundamental Rights AND other legal rights
- Prohibition & Certiorari cannot be issued against purely legislative bodies or private individuals

→ Article 21 — Landmark Cases

Case	Year	Significance
ADM Jabalpur vs Shivkant Shukla	1976	Habeas Corpus case — Art 21 suspended during Emergency; dissenting judge Justice H.R. Khanna held Right to Life is a basic human right, not a charity
Maneka Gandhi Case	1978	SC gave a wider interpretation to Art 21 — introduced US-style "Due Process of Law"

🧠 MEMORY TRICK

Article 21 textually says "**procedure established by law**" — borrowed from **Japan's Constitution**. "**Due process of law**" is a US concept — first recognised in India via the Maneka Gandhi case (1978).



②④ Ancient History — Sangam Literature & Tolkappiyam 📖

Q24. The famous Tamil grammar book Tolkappiyam, the oldest surviving work of the Sangam period, was written by whom?

- ☐ A Agathiyar
- ☒ B Tolkappiyar ✓
- ☐ C Hango-Adigal
- ☐ D Thiruvalluvar

✓ Answer: B — Tolkappiyam, the oldest surviving Tamil grammar text, was composed by Tolkappiyar (during the 2nd Sangam at Kapatapuram)

- 🔑 Sangam Age: **3rd century BC – 3rd century AD**
- 🔑 3 Sangams: 1st & 3rd at **Madurai**; 2nd at **Kapatapuram** (Tolkappiyam composed here)
- 🔑 Sangam works classified into **Mel Kanakku** (didactic) and **Kil Kanakku** (narrative — incl. Ettuthogai & Pattuppattu)
- 🔑 Genres within poetry: **Akam** = love themes; **Puram** = war themes

→ Major Sangam-Era Epics

Epic	Author	Key Story
Silappadikaram	Ilango Adigal	Kovalan, Madhavi & Kannagi; Kannagi worshipped as goddess of chastity (Pattini Cult, started by Chera king Cheran Senguttuvan)
Manimekalai	Sattanar	Story of Manimekalai, daughter of Kovalan & Madhavi

→ Sangam Politics — Chera, Chola, Pandya

Dynasty	Capital	Ports	Emblem
Chera	Vanji / Karur	Muziris (Mozari), Tondi	Bow & Arrow
Pandya	Madurai	Korkai	Fish
Chola	Uraiyur (later Gangaikonda Cholapuram, Thanjavur)	—	Tiger

SANGAM SOCIAL CLASSES (TAMIL REGION)

- Vellalar** = rich landlords (top)
- Uzhavar** = ordinary plowmen/farmers (middle)
- Adimai/Kadaiyiar** = slaves/manual workers (bottom)



25 Polity — Powers of the Governor

Q25. What is the primary role of the Governor regarding most state universities?

- ☐ A Governor has no role in state universities
- ☒ B Governor is the ex-officio Chancellor and appoints the Vice-Chancellor ✓
- ☐ C Governor only inaugurates convocation ceremonies
- ☐ D Governor decides the university curriculum

✓ Answer: B — Governor = ex-officio Chancellor of most state universities; appoints the Vice-Chancellor

→ Governor's Constitutional Discretion

Article	Power
200	May reserve a State Bill for the President's consideration
213	Power to promulgate Ordinances when the State Legislature is not in session
161	Pardoning power (compare Art 72 — President's pardoning power)

⚠ EXAM TRAP!

Unlike the President, the **Governor CANNOT pardon a death sentence**, and **CANNOT pardon a court-martial sentence** — both remain exclusive Presidential powers under Article 72.

Like a Money Bill at the Centre (needs the President's recommendation), a Money Bill in a State cannot be introduced without the **Governor's prior recommendation**.



26 Art & Culture — Bharat Mata Painting & Bengal Partition

Q26. Who painted the iconic artwork "Bharat Mata," symbolising India as a divine mother and inspiring nationalist sentiment during the freedom movement?

- ☒ A Abanindranath Tagore ✓
- ☐ B Nandalal Bose
- ☐ C Jamini Roy
- ☐ D Asit Halder

✓ Answer: A — Abanindranath Tagore painted "Bharat Mata" in 1905, during the Partition of Bengal

KEY FACTS

- 🔑 *Bharat Mata is shown with **4 hands** holding: a book, a sheaf of paddy/crops, a piece of cloth, and a garland*
- 🔑 *Abanindranath Tagore founded the **Indian Society of Oriental Art***
- 🔑 ***Nandalal Bose** — a student of the Indian Society of Oriental Art — decorated/illustrated the original Constitution of India*
- 🔑 *Constitution Calligraphy: English by **Prem Behari Narain Raizada**; Hindi by **Vasant Krishan Vaidya***

→ Anti-Partition (Swadeshi) Movement

- **1905** *Bengal Partition announced by **Curzon**; enforced October 1905; opposed via Rakhi Bandhan ceremonies for Hindu-Muslim unity*
- **1906** *Congress session at Calcutta presided by **Dadabhai Naoroji** — adopted Swaraj, Boycott, National Education & Swadeshi resolutions; **V.O. Chidambaram Pillai** formed the Swadeshi Steam Navigation Company*
- **1911** *Bengal Partition **annulled** at the 3rd Delhi Durbar (Viceroy **Hardinge II**); capital to shift to Delhi*
- **1912** *Capital formally shifted from Calcutta to Delhi*

THE THREE DELHI DURBARS

1877 — Lord Lytton (Queen Victoria proclaimed Empress of India)

1903 — Lord Curzon (Edward VII proclaimed Emperor)

1911 — Lord Hardinge II (George V; Bengal Partition annulled, capital → Delhi)



②⑦ Geography — Rocks & the Deccan Traps

Q27. The Deccan Traps region of India is primarily made up of what type of rock?

- ☐ A Granite (~~Plutonic igneous~~)
- ☒ B Basalt (Extrusive igneous) ✓
- ☐ C Marble (~~Metamorphic~~)
- ☐ D Sandstone (~~Sedimentary~~)

✓ Answer: B — Deccan Traps formed from basaltic (Flood Basalt) lava → black/regur soil typical of volcanic regions

→ Rock Cycle – Basics



Igneous (from solidification of magma/lava — the primary rock) → **Metamorphic** (via recrystallisation under heat/pressure) & **Sedimentary** (via erosion + deposition in layers)

Igneous (Original)	Becomes Metamorphic
Granite	Gneiss
Limestone (sedimentary)	Marble
Sandstone (sedimentary)	Quartzite
Shale (sedimentary)	Slate / Schist
Coal (sedimentary)	Graphite

IGNEOUS ROCKS

Plutonic/Intrusive (formed inside crust):

Granite → continental crust is granitic

Volcanic/Extrusive (formed on surface):

Basalt, Gabbro, Pumice → oceanic crust is basaltic

SEDIMENTARY & METAMORPHIC FEATURES

Sedimentary types: **organic**, **chemical**, **mechanical (clastic)** — contain fossils, form in layers

Metamorphic rocks show **foliation/lineation** (banding)

Dynamic metamorphism = metamorphism with no significant chemical change

KEY FACTS

- 🔑 Study of rocks = **Petrology**
- 🔑 Continuous conversion of one rock type into another = **Rock Cycle**
- 🔑 Continental crust = thick & light (granite/SIAL); Oceanic crust = thin but denser (basalt/SIMA)

→ Volcanism — India

Volcano Type	Nature	Example
Caldera	Highly explosive	—
Composite/Strato	Thick lava	—
Shield	Less viscous lava	Mauna Loa, Mauna Kea (Hawaii, USA)
Flood Basalt	Most fluid, least viscous lava	Deccan Traps

HOTSPOT VOLCANISM

Occurs at the **middle of a tectonic plate** (not at a plate boundary); lava believed to rise directly from the **outer core**. The **Réunion hotspot** is linked to the formation of the Deccan Traps as the Indian Plate drifted over it.

India's only active volcanoes: **Barren & Narcondam Islands** (Andaman & Nicobar)



28 Modern History — Muslim League & the Pakistan Demand 🕌

Q28. Who presided over the Allahabad Session of the Muslim League (1930)?

- ☒ A Allama Iqbal ✓
- ☐ B Muhammad Ali Jinnah
- ☐ C Aga Khan
- ☐ D Choudhry Rahmat Ali

✓ Answer: A — Allama Iqbal presided over the 1930 Allahabad Session, an early articulation of a separate Muslim state

KEY FACTS

- 🔑 **Muslim League** founded in **1906** at Dhaka by **Nawab Salimullah & the Aga Khan**
- 🔑 Official reason for Bengal Partition (1905): administrative convenience; real reason: to weaken Bengal — a nerve centre of nationalist activity — by dividing it along Hindu-Muslim lines
- 🔑 The term "**Pakistan**" was coined by **Choudhry Rahmat Ali** in **1933** (in his pamphlet "Now or Never")
- 🔑 Acronym: **Punjab + Afghania (NWFP) + Kashmir + Sindh + Baluchistan**



★ Bonus — Washington Consensus 🏦

QUICK FACT

The **Washington Consensus** = a set of free-market economic policy prescriptions (by bodies like the **IMF & World Bank**) to help developing nations overcome economic crises & hyperinflation. Coined by economist **John Williamson** in **1989**.



📋 Quick Revision — All Important Years (Lecture 11)

Event	Year
Cheetah extirpated from India	1952
Project Tiger launched	1973
Jim Corbett National Park established	1936
Madhav NP declared 58th Tiger Reserve	9 March 2025
Continental Drift Theory (Wegener)	1912
Convection Current Theory (Arthur Holmes)	1930
Plate Tectonic Theory (McKenzie/Parker/Morgan)	1967–68
Global Hunger Index 2025 — India's rank	102/123, Score 25.8
SEBI established / statutory status	1988 / 1992
SIDBI established	1990

Tangail Saree GI tag	2024
1st Amendment — 9th Schedule, Art 31A & 31B	1951
44th Amendment — Right to Property removed from FRs	1978
Kesavananda Bharati case (Basic Structure Doctrine)	1973
I.R. Coelho case	2007
Indian Councils Act (Morley-Minto)	1909
Gol Act (Montagu-Chelmsford / Dyarchy)	1919
Gol Act (Provincial Autonomy)	1935
Cabinet Mission Plan	1946
Objectives Resolution moved / adopted	13 Dec 1946 / 22 Jan 1947
Constitution adopted / came into force	26 Nov 1949 / 26 Jan 1950
ADM Jabalpur case	1976
Maneka Gandhi case	1978
Bharat Mata painting by Abanindranath Tagore	1905
Bengal Partition announced / annulled	1905 / 1911
Muslim League founded	1906
"Pakistan" term coined (Rahmat Ali)	1933
Washington Consensus	1989



Topper Tips — Exam Ke Liye Yaad Rakho!

EXAM TIP

1. **Narrow Money = M1**, **Broad Money = M3** — don't mix them up
2. **Governor CANNOT pardon death sentence or court-martial** — exclusive to the President
3. Article 21 wording ("procedure established by law") = from **Japan**; "Due process" concept = from the **USA**
4. Union Article + **89** = corresponding State Article

EXAM TIP

5. Project Tiger (1973) ≠ Cheetah Reintroduction Programme (Kuno NP) — don't confuse the two schemes
6. Quad = informal grouping, **NOT** a treaty-based alliance like NATO
7. India's K2 (8,611 m) is technically India's highest peak, but lies in **PoK**
8. Oligosaccharide = **2-10** monosaccharide units (not "10+", that's polysaccharide)

Note: This transcript contained several speech-recognition errors and a couple of self-corrected slips by the instructor (e.g., Project Cheetah vs Project Tiger, Bengal Partition "annulled" vs "annual," Arthur Holmes vs "Arthur Romes"). Facts above have been cross-checked and corrected for accuracy wherever the spoken content was ambiguous or contradictory – current-affairs figures (GHI 2025, Madhav Tiger Reserve, SEBI Chairman, Tangail GI tag) were verified against the latest available sources.



END OF LECTURE 11 NOTES

All 28 Questions Covered  | Practice karo, Mock do, Top karo! 